



Internship Report of the 2nd Year

**Modification of the Eurotop overtopping equations for
breakwaters during their construction.**

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Scientific and Academic integrity

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Abstract

Most overtopping equations are based on the geometry of the structure and the wave conditions, but they do not take into account the construction phase of the breakwater. During this phase, the geometry of the breakwater changes considerably, and the physical assumptions underlying the equations are no longer valid. In this study, a well-calibrated Volume of Fluid (VOF) model is employed to compare simulation results with those obtained from the IH2VOF model. The test cases are based on the construction of a rubble mound breakwater with a 1:1.5 slope. The overtopping discharge is measured for each construction stage. Finally, the proposed corrections are validated against a blind test case involving different geometries and wave conditions.

Keywords: overtopping, construction phase, VOF model, mean overtopping discharge, rubble mound breakwater, EurOtop, IH2VOF

1. Acknowledgements

I would like to express my sincere gratitude to my supervisor, Mr. Gabriel Diaz Hernandez, who provided invaluable guidance and support throughout the course of this project. His expertise and insights have been instrumental in shaping the direction of my research and ensuring its success.

2. Introduction

2.1. Motivation

Predicting overtopping discharge is important for the design of coastal structures because overtopping can damage the structure and endanger the people it protects. The consequences of a storm during the construction phase of a breakwater can be serious, not only because of the risks to people working on site and to the equipment, but also because a storm can destroy the breakwater during construction. Consequently, companies could save substantial amounts of money by adapting construction schedules to wave forecasts rather than rebuilding the breakwater after a storm. Moreover, the construction phase of a breakwater can last a long time, and its geometry

changes considerably during this phase. This can lead to an increase in overtopping discharge and, consequently, to a higher risk of structural damage and danger to people working on site.

2.2. State of the art

2.2.1. Wave Overtopping Prediction Methods

The main method for predicting overtopping discharge is the EurOtop methodology [1], which provides a set of equations for a wide range of coastal structures. These equations are easy to use: they require the wave parameters, the structure height, and the structure type, and directly provide the overtopping discharge. The equations in the EurOtop manual are empirical and were derived from analyses of a large database, primarily the CLASH database [2]. This database contains more than 10,000 overtopping tests conducted on different types of structures and under different wave conditions. The tests were conducted in many European countries, mainly in laboratories at reduced scale, although some were carried out in the field. EurOtop is an improvement on CLASH, as it is designed for full-scale breakwaters and can address more complex structures.

New neural-network-based overtopping predictors have been developed in recent years [3]. These predictors rely on neural networks trained on large databases of overtopping data, such as CLASH. These predictors provide good results. Their main difference from the EurOtop equations is that they are not based on physical assumptions but on statistical analyses of the data. This means that they can be used for more complex structures and wave conditions, but they are less reliable for extrapolation.

2.2.2. Overtopping of Rubble Mound Breakwaters

A rubble mound breakwater is composed of three major parts: the core, the underlayers, and the armour layer. Each of these parts has physical properties that allow it to dissipate varying amounts of energy. The roughness of the armour layer plays a significant role in energy dissipation during overtopping. Indeed, the rougher the armour layer, the more energy is dissipated through friction. This roughness is complemented by porosity, which also contributes to energy dissipation. A perfectly impermeable slope dissipates much less energy than a porous structure. Moreover, the crest freeboard has a significant impact on overtopping discharge: the higher the freeboard, the lower the overtopping discharge. This is because the freeboard acts as a barrier to the waves, allowing them to dissipate more energy on the structure

before overtopping. The geometry of the breakwater also plays a role in overtopping discharge. A steeper slope leads to a higher overtopping discharge than a gentler slope because the waves have a shorter distance over which to dissipate energy.

2.2.3. Temporary and Transitional Coastal Structures

The construction phase of a breakwater can be compared to a temporary coastal structure, such as a gabion-box artificial reef or a geotextile sand container (GSC).

The hydraulic performance of these temporary structures is not well known, as they are not designed to be permanent. The overtopping discharge from these structures can be very high because they are not designed to dissipate energy. During construction, the breakwater can be considered a temporary structure, and there are no equations for predicting its overtopping performance, despite the significant practical importance of this problem in coastal engineering projects.

2.3. Objectives

The main objective of this project is to modify the EurOtop equations to account for the construction phases. To achieve this, the IH2VOF model is used as a numerical tool developed at IHCantabria that applies the Volume of Fluid (VOF) method to simulate wave interactions with coastal structures. A set of simulations was conducted for multiple construction stages of a rubble mound breakwater.

The aim was to determine correction factors for the EurOtop equations that account for the construction phases. These coefficients were linked to the physical properties of the breakwater during each construction phase. This makes it possible to determine the correction factors using a set of parameters related to crest freeboard, the degree of layer completion, and the apparent roughness and porosity of the breakwater. To represent the different construction stages, a parameter representing the construction stage in terms of its influence on overtopping discharge was created. This parameter is linked to the physical properties of the breakwater and is used to determine the correction factor for each construction stage. This makes it possible to propose a practical engineering formulation suitable for design and construction planning.

3. Context of the internship

3.1. IHCantabria

3.1.1. *Their research*

The IHCantabria (Institute of Hydraulics of Cantabria) is a public research institute located in Santander, Spain. It comprises 14 research groups and published 75 papers in 2025, of which 27 were classified as D1 and 29 as Q1. Its research focuses on environmental hydraulics, including coastal and harbour engineering, offshore and marine energy. The institute also conducts research on oceanography and the impact of climate change on ecosystems.

3.1.2. *Partners and collaborations*

The IHCantabria is a public research institute that is part of the University of Cantabria. It collaborates with 23 public administrations in Spain and Europe, three research centres, and two universities. It also collaborates with private companies: 12 in the energy sector, eight in engineering, and four in construction.

3.1.3. *Their financial support*

IHCantabria is a public research institute funded by the Spanish government and the European Union. It also collaborates with private companies that provide financial support for specific projects. These collaborations allow the institute to be 90% self-financed and to manage a budget of €12.5 million in 2025. This budget is allocated 47% to research and 53% to private contracts and consulting.

3.2. *My experience*

I worked on the development of a new methodology for predicting overtopping discharge during the construction phase of a rubble mound breakwater. I worked largely independently under the supervision of Mr. Gabriel Diaz Hernandez. I was given considerable freedom to develop the methodology and determine the best way to achieve the project objectives. I was also given considerable freedom to choose suitable tools and methods. I conducted extensive research into different mathematical and statistical approaches to identify the best way to achieve these objectives. I also carried out numerous simulations using the IH2VOF model to validate the methodology. Several approaches did not produce the expected results, but this was a valuable opportunity to learn from mistakes and identify the most effective method.

The following sections describe the final methodology and the results obtained. This methodology is the result of extensive research and simulations and represents the most effective approach identified in this project.

4. Methodology

4.1. Case study Breakwater

To conduct a reliable study, it is necessary to use a prototype breakwater with precisely defined dimensions and proportions. The assumptions concerning these proportions are as follows:

1. The breakwater is composed of four parts: the core, the armour layer, Underlayer 1, and Underlayer 2.
2. The breakwater has a constant slope from the toe to the crest.
3. The layers have a constant width from the toe to the crest of the breakwater.
4. All other parts, such as the crown wall, the dam, and the toe, are not considered in the model, as they do not have a significant influence on overtopping discharge.

The prototype used in this study to develop correction factors had a 1:1.5 slope. The core is made of small stones, the underlayers are made of larger stones, and the armour layer is made of very large stones. The armour layer is designed to dissipate energy during overtopping, while the underlayers support the armour layer and also dissipate energy. The core supports the entire structure and also dissipates energy. The breakwater has a crest freeboard of $7.50m$ and a crest height of $15m$.

This section is inspired by Tramo 1 of the breakwater in the Port of Alicante, Spain. See Appendix Appendix A.

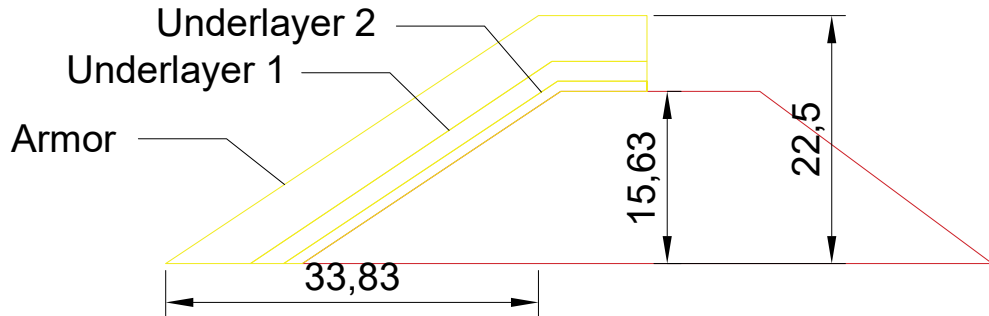


Figure 1: Complete breakwater

The dimensions are as follows.

Quantity	Value (m)
Still water level	15
Crest height	22.50
Freeboard R_c	7.50

Table 1: Height and levels of the breakwater.

4.2. Definition of construction phases

As a first approach, each part of the breakwater was reduced by half in each simulation, resulting in 3×2 configurations for the calibration case. The armour layer was removed first; once it had been completely removed, Underlayer 1 was removed, followed by Underlayer 2. This resulted in a total of six configurations representing the different construction stages. The first stage corresponds to the complete breakwater, the second to the breakwater

without the armour layer, and the third to the breakwater without the armour layer or Underlayer 1. This sequence continued until the final stage, in which no layers remained.

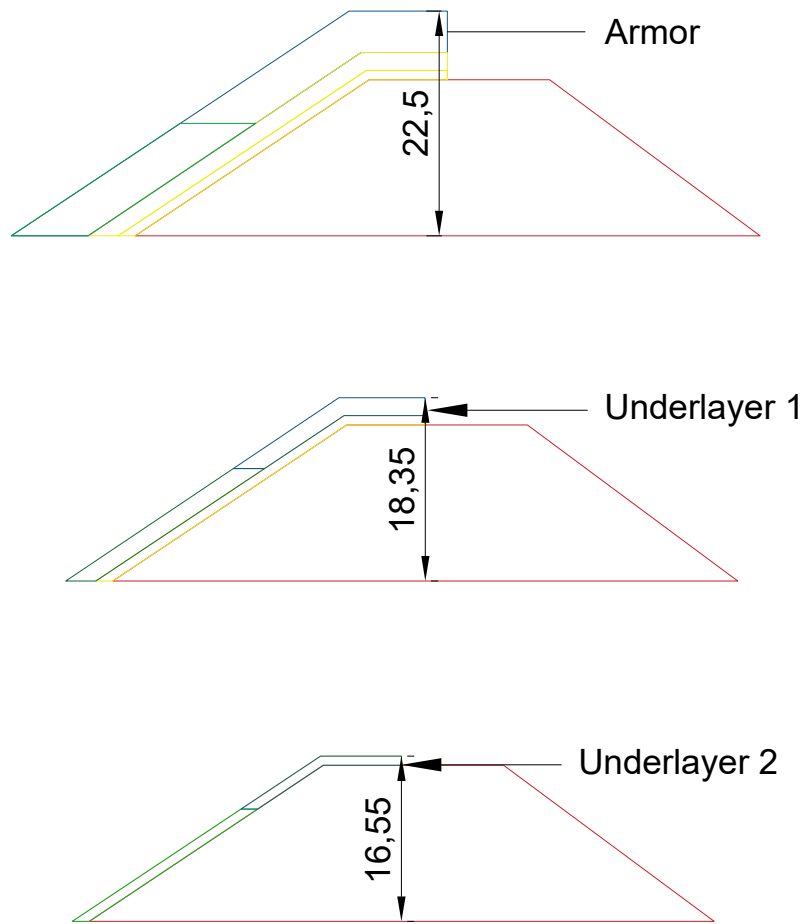


Figure 2: Phases of the construction

4.3. Numerical model

4.3.1. IH2VOF model

The fluid domain is governed by the two-dimensional Reynolds-Averaged Navier-Stokes (RANS) equations. The pressure and velocity fields are decomposed into mean and fluctuating components. The RANS equations are formulated as follows:

$$\frac{\partial \bar{u}_i}{\partial t} + \bar{u}_j \frac{\partial \bar{u}_i}{\partial x_j} = -\frac{1}{\rho} \frac{\partial \bar{p}}{\partial x_i} + g_i + \frac{1}{\rho} \frac{\partial \bar{\tau}_{ij}}{\partial x_j} - \frac{\partial \overline{u'_i u'_j}}{\partial x_j} \quad (1)$$

where $\bar{u}_i$ represents the mean velocity, $\bar{p}$ the mean pressure, g_i the gravitational acceleration, $\bar{\tau}_{ij}$ the mean viscous stress tensor, and $\overline{u'_i u'_j}$ the Reynolds stress tensor. The Reynolds stress tensor is modelled using a turbulence model; in this case, the k- ϵ SST model.

The standard RANS equations are strictly valid only for fluid domains without internal obstacles. To model porous media, the governing equations are augmented with momentum sink terms that account for energy losses caused by solid obstructions. The VARANS equations used in the IH2VOF model are expressed as follows:

$$\left(\frac{1 + C_A}{n} \right) \frac{\partial \bar{u}_i}{\partial t} + \frac{\bar{u}_j}{n^2} \frac{\partial \bar{u}_i}{\partial x_j} = -\frac{1}{\rho} \frac{\partial \bar{p}}{\partial x_i} + g_i + \frac{\nu}{n} \frac{\partial^2 \bar{u}_i}{\partial x_j \partial x_j} - \frac{1}{n^2} \frac{\partial \overline{u'_i u'_j}}{\partial x_j} - \frac{\alpha \nu (1 - n)^2}{n^3 D_{50}^2} \bar{u}_i - \frac{\beta (1 - n)}{n^3 D_{50}} \bar{u}_i |\vec{u}| \quad (2)$$

4.3.2. Numerical Domain

A structured mesh with a uniform grid is used around the breakwater and a stretched grid is used in the rest of the domain. The mesh is refined around the breakwater to accurately capture the flow features. Around the breakwater, the grid spacing is $0.5m \times 0.5m$. Away from the breakwater, the grid expands to a maximum spacing of approximately $1.5m$. The cell sizes are shown in the following figure:

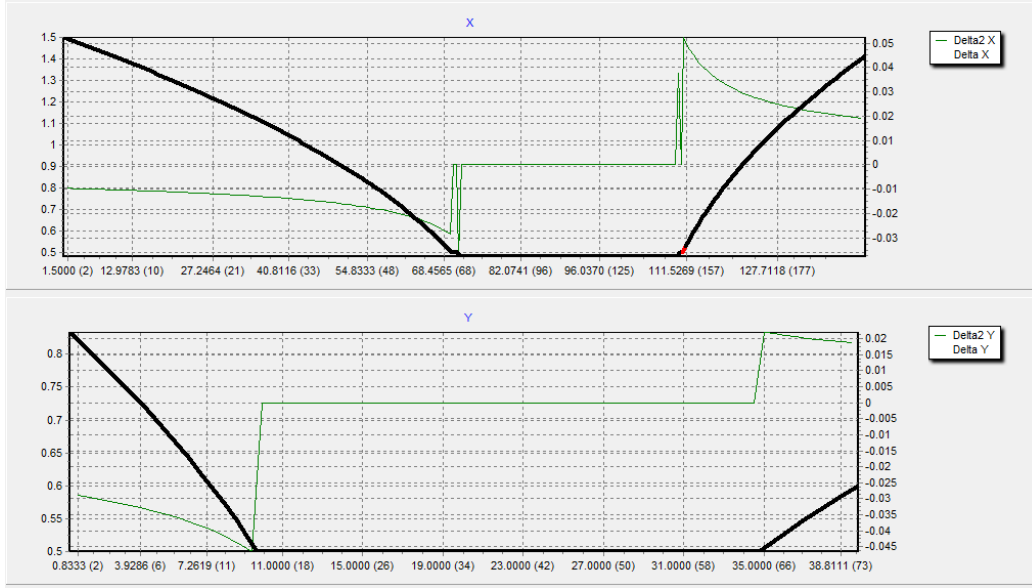


Figure 3: Computational mesh

Four wave gauges are used: three are placed before the breakwater to measure the incident wave height, and one is placed at the top of the breakwater to measure overtopping discharge.

4.3.3. Boundary conditions

The left boundary of the simulation domain, where the wave paddle is located, is configured to absorb the waves, thereby preventing reflection from the left boundary. The right boundary is not configured to absorb waves because the waves do not reach it due to the presence of the breakwater. The water level is kept constant at $15m$ for all simulations of the first section, allowing the results for the different construction stages to be compared under the same water-level conditions. Waves are generated using a piston-type wavemaker, which is a common method for generating waves in numerical simulations.

4.4. Wave conditions

To generate only relevant waves, a DoE matrix was developed to select the wave conditions relevant to the breakwater. The matrix is based on wave height and wave steepness. Wave height is defined as the significant wave height H_{m0} , while the wave period, defined as the spectral mean period

$T_{m-1,0}$, is derived from the wave steepness. The DoE matrix includes 10 wave-height levels and five wave-steepness levels, resulting in a total of 50 simulations for each construction stage. These levels were selected to cover a wide range of wave conditions, from low to high waves and from short to long waves. They were based on the design wave conditions for the breakwater, with values ranging from $H_{m0} = 2m$ to $H_{m0} = 12m$ and from $s_{m-1,0} = 0.015$ to $s_{m-1,0} = 0.070$.

The main objective of this matrix is to cover a wide variety of wave conditions while ensuring adequate coverage of the conditions relevant to the breakwater. For each wave condition, the Iribarren number and the critical wave height are calculated to determine whether the condition is relevant to the breakwater. The critical wave-height ratio was defined as $H_{m0}/h > 0.45$, and the critical Iribarren number was defined as $xi_{m-1,0} < 2.50$. To save computational time, wave conditions that are not relevant to the breakwater were excluded from the simulations. Of the 266 possible simulations, only 105 were valid after applying these criteria, which resulted in a substantial reduction in computational time.

4.5. Overtopping Measurement Procedure

To measure overtopping discharge, a gauge located at the abscissa of the breakwater crest is used. This gauge provides the water velocity through each cell in the column at each time step. The velocity is integrated over the portion of the mesh above the breakwater crest to calculate the volume of water passing over the breakwater at each time step. This volume is then integrated over the entire simulation to obtain the total volume of water passing over the breakwater. Finally, the total volume is divided by the simulation duration to obtain the overtopping discharge in m^3/s . The calculation follows the formula below:

$$q = \frac{1}{T_{\text{sim}}} \sum_{j=1}^{N_t} \left(\sum_{i=\text{crest}}^{\text{top}} u_{i,j} \cdot \Delta z_i \right) \Delta t_j \quad (3)$$

4.6. Determination of Incident wave conditions

The two-gauge method [4] is used to calculate the spectral wave height and period from the spectra measured at each gauge. The first gauge is placed 10m before the toe of the breakwater, and the second is placed 30m before the first gauge. The MATLAB `pwelch` function is used to calculate

the spectrum at each gauge. The separation of incident and reflected waves is then used to calculate the incident-wave spectrum. The formula is as follows:

$$S_{inc}(f) = \frac{S_{11}(f) + S_{22}(f) - 2 \cdot \Re(S_{12}(f) \cdot e^{+ik\Delta x})}{4 \cdot \sin^2(k\Delta x)} \quad (4)$$

The incident-wave spectrum is then integrated to calculate the spectral moments m_0 and $m_{-1,0}$. These moments are used to calculate the significant wave height and spectral period using the following formulas:

$$H_{m0} = 4\sqrt{m_0} \quad (5)$$

$$T_{m-1,0} = \sqrt{\frac{m_{-1}}{m_0}} \quad (6)$$

5. Model validation

5.1. Validation Against EurOtop Smooth-Slope Equations

The simulation results were compared with the EurOtop equations. As a first step, the model was validated using Equation 5.10 from the EurOtop manual, which is valid only for smooth slopes. This equation was used to calibrate the model and the overtopping-discharge measurement method. It is expressed as follows:

$$\frac{q}{\sqrt{g} H_{m0}^3} = \frac{0.023}{\sqrt{\tan \alpha}} \cdot \gamma_b \cdot \xi_{m-1,0} \cdot \exp \left(- \left[2.7 \cdot \frac{R_c}{H_{m0} \cdot \xi_{m-1,0} \cdot \gamma_f \cdot \gamma_\beta \cdot \gamma_v} \right]^{1.3} \right) \quad (7)$$

Where α represent the slope of the breakwater, R_c the crest freeboard, γ_b the berm correction factor, γ_f the friction correction factor, γ_β the oblique wave incidence correction factor, and γ_v the vertical wall correction factor. The parameters γ_b , γ_β , and γ_v are equal to 1 because there is no berm, the wave incidence angle is zero, and there is no vertical wall. The friction parameter γ_f is equal to 1 for a smooth slope. A simulation with a smooth, impermeable slope inclined at 1:1.5 was performed, and the overtopping discharge was measured using the method described in the previous section. The simulation results were then compared with the results obtained using EurOtop Equation 5.10. The results are shown in the following figure.

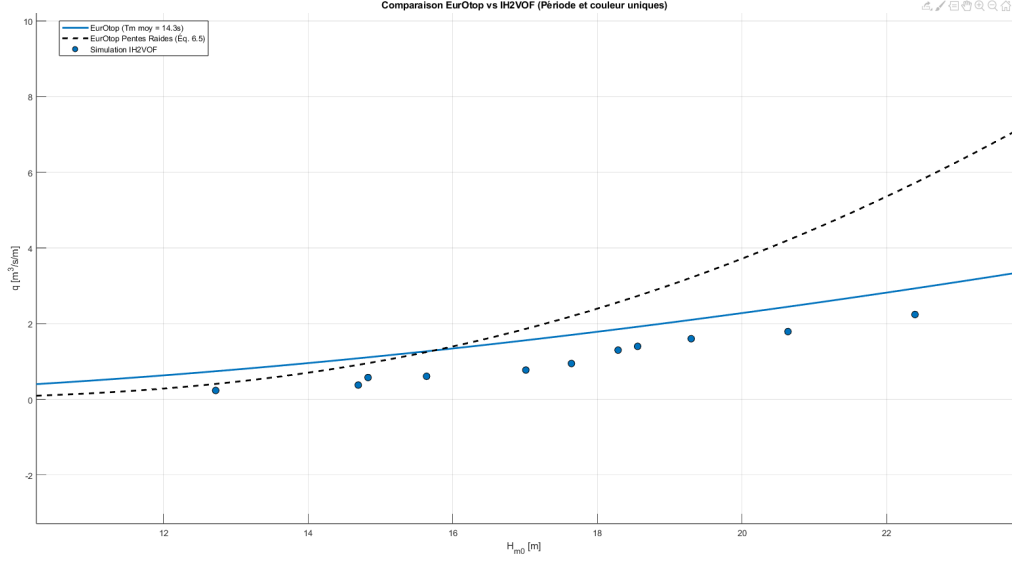


Figure 4: Calibration of the model with the EurOtop equations for a smooth slope.

The simulation results are in good agreement with the results of EurOtop Equation 5.10, although EurOtop slightly underestimates the overtopping discharge. However, the agreement is sufficiently good to validate both the model and the overtopping-discharge measurement method.

5.2. Validation Against EurOtop Armoured-Slope Equations

Following the satisfactory results obtained with Equation 5.10, Equation 6.5 was applied using the same methodology. Since it is valid for armoured slopes, Equation 6.5 is expressed as follows:

$$\frac{q}{\sqrt{g \cdot H_{m0}^3}} = 0.09 \cdot \exp\left[-\left(1.5 \cdot \frac{R_c}{H_{m0} \gamma_f \gamma_\beta}\right)^{1.3}\right] \quad (8)$$

Because the IH2VOF simulations are two-dimensional, the parameter γ_β is equal to 1 because the wave incidence angle is zero. The parameter γ_f is equal to 0.45 for this type of armor as specified in the EurOtop manual. This formulation shows good agreement between the model results and the EurOtop equations for the fully constructed breakwater.

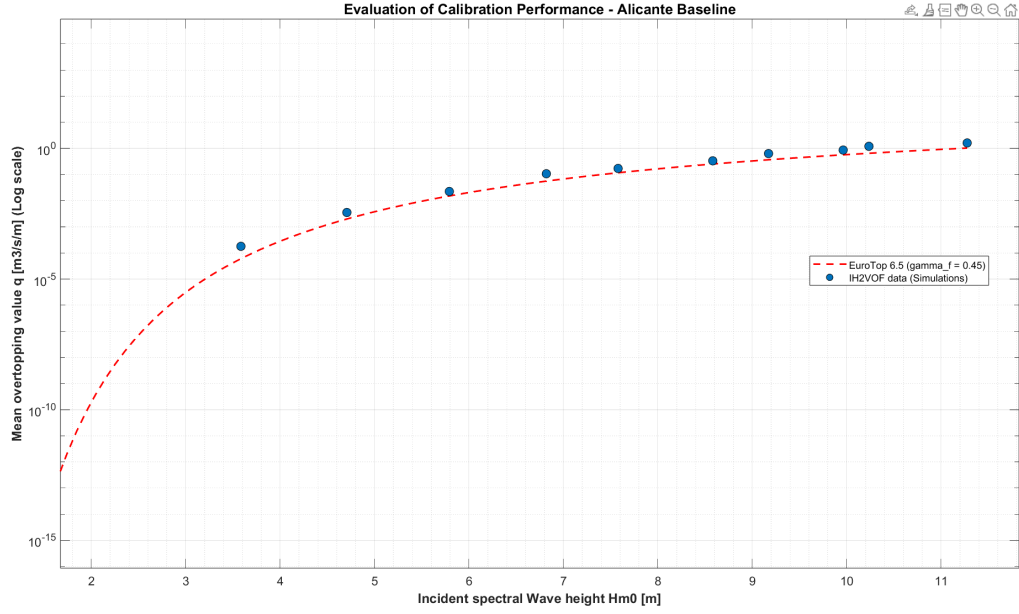


Figure 5: Calibration of the model with the EurOtop equations.

6. Overtopping behaviour during the construction phase

The following figure presents the simulation results for the different construction stages of the breakwater. The solid curve represents the results of the EurOtop equation, while the dots represent the simulation results.

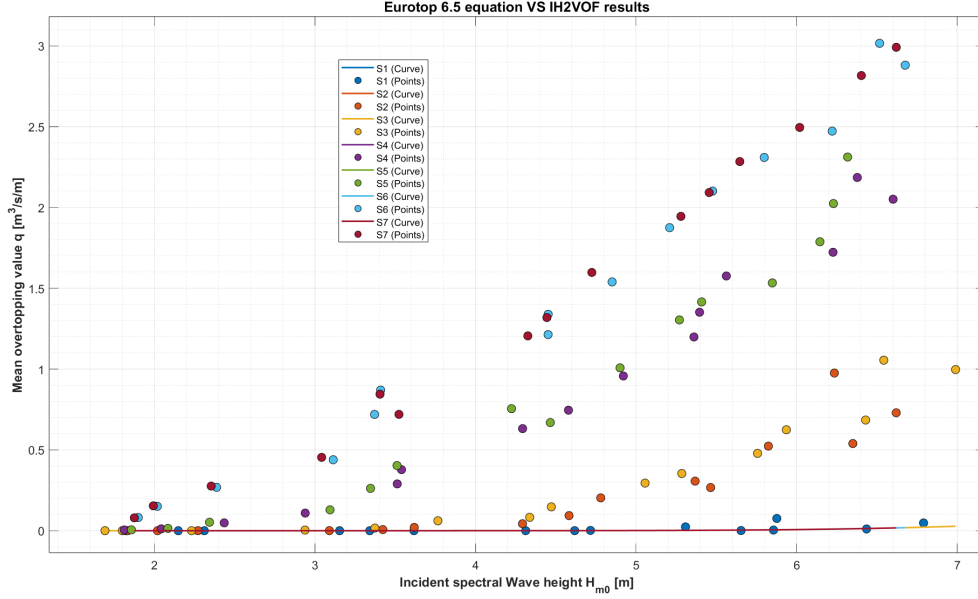


Figure 6: First results of the simulations for the different stages of the construction. The stages of construction are marked S1 to S7, the more S is high, the less the breakwater is complete.

The results show that overtopping increases as parts of the breakwater are removed, as expected because the crest freeboard R_c decreases. Moreover, the friction parameter is no longer valid because the waves expose not only the armour layer but also the underlayers, which were not designed to be directly exposed to wave action.

7. Development of Construction Correction Factors

7.1. Construction Completion Index

To improve the prediction of overtopping discharge during the construction phase, a new parameter called the Construction Completion Index I_h is introduced. This index quantifies the degree of completion of the breakwater and is determined by a weighted linear combination of the construction or degradation state of each layer:

$$I_h = w_R I_R + w_A I_A + w_{U1} I_{U1} + w_{U2} I_{U2} \quad (9)$$

This index ranges from 0 (structure absent) to 1 (structure intact).

Definition of Terms

- w_R, w_A, w_{U1}, w_{U2} : Weighting factors for the crest freeboard, armour layer, Underlayer 1, and Underlayer 2, respectively, where $\sum w_i = 1$.
- I_R, I_A, I_{U1}, I_{U2} : Normalized construction or integrity states of each layer.

The weighting factors are determined according to the relative influence of each layer on overtopping discharge. They are calibrated in the next section to achieve the best agreement between the model results and the EurOtop equations.

7.2. Alternative Correction Formulations

The mean overtopping discharge q (in $\text{m}^3/\text{s}/\text{m}$) is calculated using a modified version of the EurOtop equation that accounts for the new R_c , the new friction parameter, and the new breakwater geometry. A structural correction coefficient K_c is introduced to account for the effects of the sublayers on overtopping discharge. The modified equation is expressed as follows:

$$q = \sqrt{gH_{m0}^3} \cdot 0.09 \cdot \exp\left(-\left(\frac{1.5R_c}{H_{m0}\gamma_f K_c}\right)^{1.3}\right) \quad (10)$$

The coefficient K_c is defined as a function of the Construction Completion Index I_h calculated in the previous section:

$$K_c = 1 + \alpha(1 - I_h)^\beta \quad (11)$$

Definition of Terms

- K_c : Structural correction coefficient that accounts for the effects of the sublayers.
- I_h : Construction Completion Index calculated in the previous section.
- α, β : Empirical model-calibration parameters.

The parameters α and β are determined through a calibration process using the simulation results for different construction stages. The objective is to minimize the discrepancy between the predicted and simulated overtopping discharges across all construction phases.

7.3. Optimization Procedure

The first step of the optimization procedure is to define the weighting factors for each layer of the breakwater. It is not possible to calibrate these coefficients directly by comparing the simulation results with the EurOtop equations because there are too many parameters to calibrate. The parameters α and β must first be eliminated. This is achieved by using the simulation results to calculate the optimal value of K_c for each construction stage through a logarithmic least-squares method. The optimal value of K_c is obtained by minimizing the following objective function:

$$\text{Objective Function} = \sum_{i=1}^N (\log(q_{sim,i}) - \log(q_{pred,i}))^2 \quad (12)$$

where $q_{sim,i}$ is the overtopping discharge obtained from the simulation for stage i , and $q_{pred,i}$ is the overtopping discharge predicted using the modified EurOtop equation with the current value of K_c . The logarithmic transformation gives equal weight to relative errors across different overtopping-discharge magnitudes.

The optimal values of K_c are then used to calibrate the weighting factors of each breakwater layer. The expression for K_c can be written as follows:

$$\ln(K_c - 1) = \ln(\alpha) + \beta \ln(1 - I_h) \quad (13)$$

Then, $\ln(K_c - 1)$ is considered as a function of $\ln(1 - I_h)$. An algorithm is used to determine the weighting factors that make the values of $\ln(K_c - 1)$ as close as possible to a linear function of $\ln(1 - I_h)$. Several initial sets of weighting factors are used to ensure convergence. Once the linear relationship has been obtained, the values of α and β are determined by performing a linear regression of $\ln(K_c - 1)$ against $\ln(1 - I_h)$. The results are shown in the following figure:

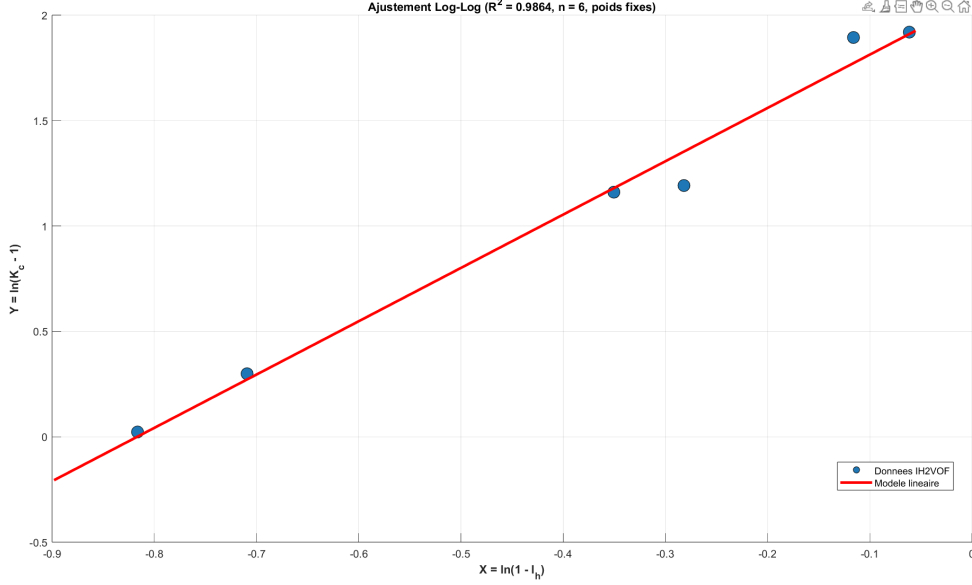


Figure 7: Linear regression of $\ln(K_c - 1)$ as a function of $\ln(1 - I_h)$.

For the first section, the following weighting factors were obtained:

Table 2: Calibration Coefficients and Weighting Factors

Parameter	Section 1
α (Calibration constant)	7.36
β (Calibration exponent)	2.71
w_R (Crest freeboard weight)	0.70
w_A (Armor layer weight)	0.15
w_{U1} (Underlayer 1 weight)	0.10
w_{U2} (Underlayer 2 weight)	0.05

It can be observed that the crest freeboard is the most influential parameter for overtopping discharge, as expected because it is the first barrier encountered by the waves. The armour layer is also important, although less influential than the crest freeboard. The underlayers are less influential but still affect overtopping discharge. These values are used to predict overtopping discharge for the different construction stages and to compare the predictions with the simulation results. The results are shown in the following figure:

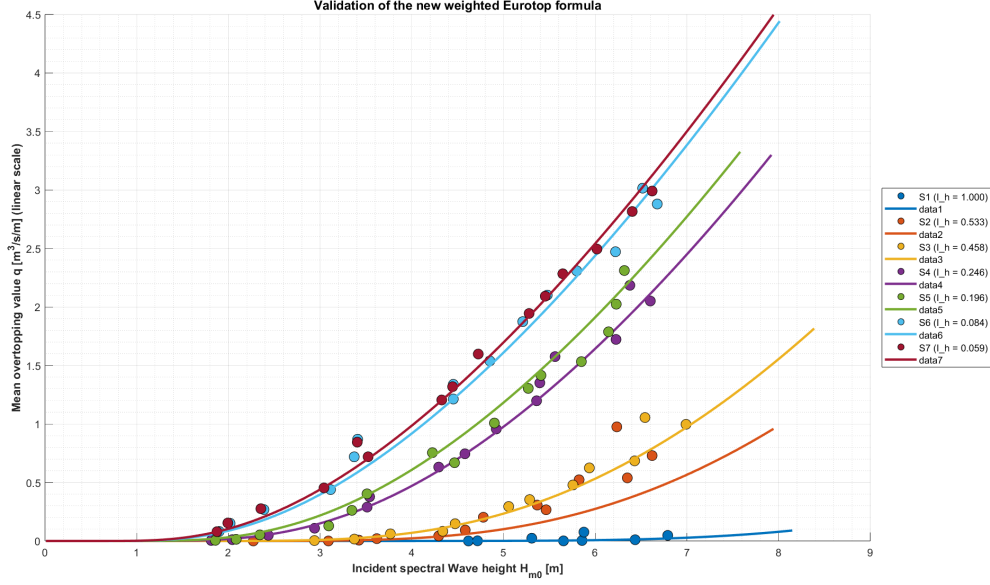


Figure 8: Comparison of predicted overtopping discharge with simulation results for different construction stages.

The results of the modified EurOtop equation with the new correction factors are in good agreement with the simulation results for all construction stages. The RMSE is 12%, which is a good result for this type of prediction.

8. Independent Validation

8.1. Partially modified section

8.1.1. Description of the modifications

To validate the new correction factors, several modifications were made to the breakwater geometry to represent alternatives to the original breakwater. The modifications were as follows:

- The slope was varied between $\cot(\alpha) = 1.88$ and $\cot(\alpha) = 3$.
- The crest freeboard was increased to $9m$ and decreased to $5m$.
- Combined modifications were made by combining the different crest-freeboard and slope modifications.

Using these modifications and the same method as for the first breakwater, together with the DoE matrix, random waves were generated to represent the range of conditions that the breakwater might encounter in practice. These modified breakwaters were simulated for randomly selected construction states, and the overtopping discharge was measured using the same method as for the first breakwater.

8.1.2. Modified slope

The results of the simulations with the modified slope were compared with the results of the modified EurOtop equation using the correction factors calculated in the previous section. The results are shown in the following figure.

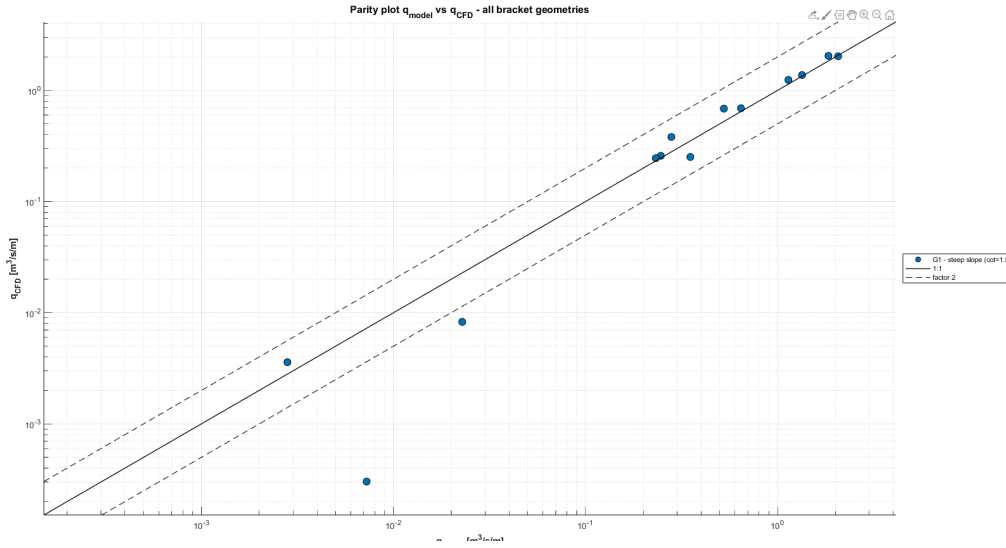


Figure 9: Modified slope compared to the calibrated EurOtop equation.

It can be observed that the tests conducted with a different slope agree very well with the predictions. Except for the lowest overtopping discharges, the simulated overtopping discharges follow the values predicted by the modified equation.

8.1.3. Higher R_c

The simulation results for the modified R_c were compared with the results of the modified EurOtop equation using the correction factors calculated in

the previous section. For this structure, fewer waves were valid because they did not satisfy the EurOtop assumptions and broke on the structure. Consequently, fewer simulations could be performed. The results are shown in the following figure.

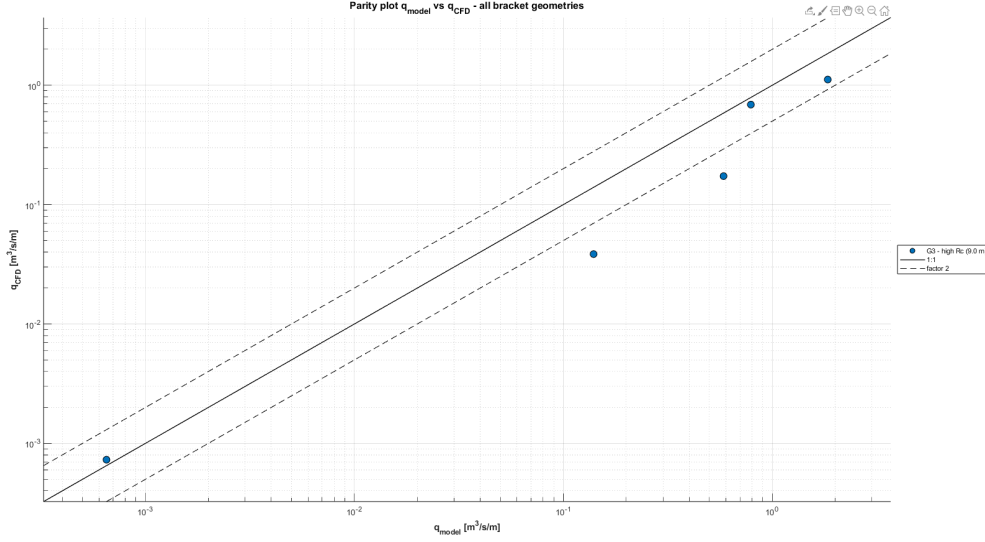


Figure 10: Higher R_c structure compared to the calibrated EurOtop equation.

The structure with the higher R_c agrees with the predictions only for the largest overtopping discharges. This can be explained by the fact that overtopping occurs as a sequence of finite volumes rather than as a continuous flow. Thus, during a one-hour simulation, one overtopping volume may be missed because of the random wave generation. If the total overtopping volume predicted by the EurOtop equation corresponds, for example, to two overtopping waves and one is missed in the simulation, the measured total overtopping volume is approximately halved. Small overtopping discharges are more affected by this phenomenon because overtopping events are less frequent. For the higher R_c , overtopping is also much less frequent than for the original R_c .

8.1.4. Lower R_c

The simulation results for the modified R_c were compared with the results of the modified EurOtop equation using the correction factors calculated in the previous section. The results are shown in the following figure.

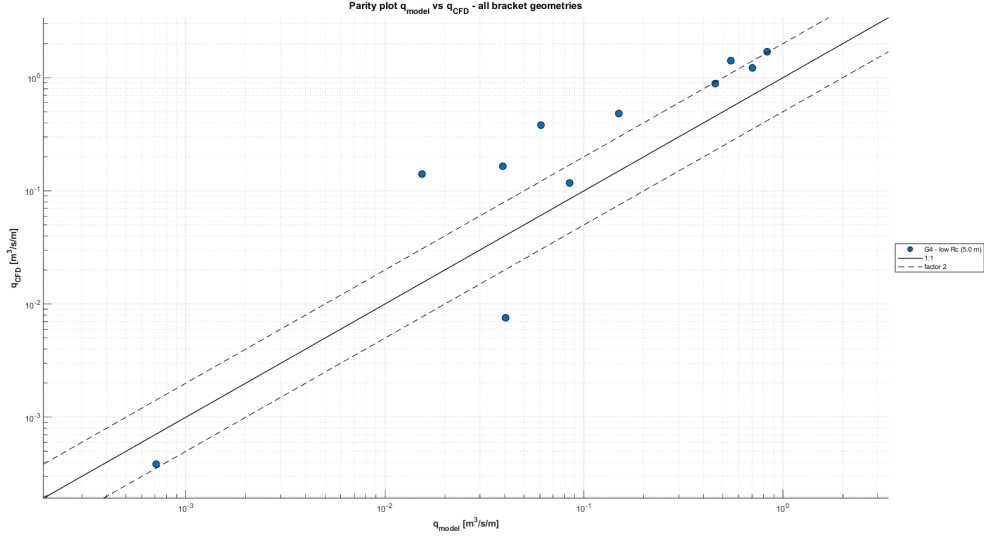


Figure 11: Lower R_c structure compared to the calibrated EurOtop equation.

The structure with the lower R_c shows the poorest agreement. However, the same phenomenon as for the higher- R_c structure can be observed. The predicted overtopping tends to match the simulation results as the wave height increases. Overtopping is much more frequent for this structure, making the flow more continuous and the total overtopping volume less sensitive to the random wave generation. Consequently, the overtopping discharge is more reliable for this structure than for the higher- R_c structure. This highlights the need to improve the model for structures with a lower R_c .

8.2. Independent Geometry

To test the newly developed correction factors on a real breakwater, a real breakwater section was selected. This section, known as Tramo 6, belongs to the breakwater at Langosteira Harbour in Spain. It was selected because several tragic events occurred during its construction, injuring workers and even resulting in the deaths of two people. The breakwater is nearly 3 km long. A schematic of the actual Section 6 can be found in Appendix Appendix B. The section used here is a simplified version of the real breakwater, based on the same assumptions as those used for the first breakwater.

The wave conditions were selected to represent the range of conditions that the breakwater might encounter in practice. A peak period of $T_p = 16s$ and

a significant wave height $H_s \in [15; 25m]$ were used.

The simulation results were compared with the results of the modified EurOtop equation using the correction factors calculated in the previous section. The results are shown in the following figure:

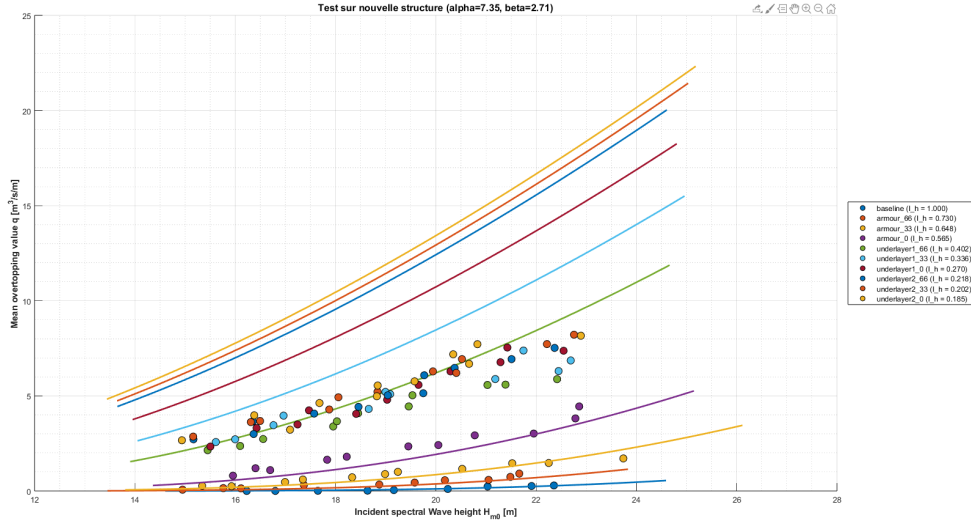


Figure 12: Comparison of predicted overtopping discharge with simulation results for the independent breakwater geometry.

The results are not as good as those obtained for the previous breakwater. The simulation results appear to reach a plateau at the highest overtopping discharges, whereas the modified EurOtop equation predicts a continuous increase. This case is too extreme for the modified EurOtop equation, and the correction factors are no longer valid. However, during the final construction stages, when the armour layer is present and overtopping is not too severe, the results agree well with the modified EurOtop equation. This shows that the correction factors developed here can improve predictions for structures of a similar scale to those used for calibration, but also that they have limitations and should be investigated further for extreme cases.

8.3. Predictive Performance

Table 3: Statistical performance comparison between original and modified EurOtop equations

Indicator	Original ($K_c = 1$)	Modified (Calib.)	Unit
<i>[Linear Scale]</i>			
RMSE	1.19664	0.12686	$\text{m}^3/\text{s}/\text{m}$
Bias	-0.83644	-0.03293	$\text{m}^3/\text{s}/\text{m}$
Scatter Index (SI)	1.4240	0.1510	-
R^2	-0.9425	0.9782	-
<i>[Logarithmic Scale & Robust Metrics]</i>			
RMSE Log	4.6941	0.6095	-
Bias Log	-3.8142	-0.0791	-
R^2 Log	-20.6188	0.9313	-
Geo. mean ratio	0.0002	0.9239	-
Geo. Scatter Index	49444.7097	1.8361	-

The predictive performance of the modified EurOtop equation with the newly developed correction factors is significantly better than that of the original formulation. The RMSE is substantially lower, indicating a much closer fit to the simulation results. The bias is also considerably lower, suggesting that the modified equation provides a more accurate estimate of the central tendency of overtopping discharge.

9. Estimation of the Maximum Overtopping Volume

9.1. Why $V_{\max}$ Must Be Considered in Addition to q

The mean overtopping discharge q (in $\text{m}^3\text{s}^{-1}\text{m}^{-1}$) is averaged over the duration of the sea state, whereas overtopping actually occurs as a sequence of discrete, highly irregular events. Only a fraction of the incident waves overtops the crest, with each wave contributing an individual volume V_i . A structure may therefore exhibit an acceptable value of q while still being subjected, during a single isolated event, to an individual volume $V_{\max}$ far exceeding what the mean discharge would suggest.

This distinction is critical for structural performance: extreme individual volumes, rather than the mean discharge, govern the mobilisation of crest and rear-slope armour units, impulsive loading on crown walls and parapets, and risks to pedestrians, vehicles, and property in the overtopping path. This is why EurOtop Section 4.2.4 requires the overtopping-volume distribution

to be statistically characterised and $V_{\max}$ to be extracted from it, in addition to, and not as a replacement for, the mean-discharge criterion.

9.2. Weibull Distribution of Individual Volumes

Following EurOtop Section 4.5, individual overtopping volumes are made dimensionless using the incident significant wave height H_{m0} :

$$V^* = \frac{V}{H_{m0}^3}, \quad (14)$$

which allows observations from different sea states, and hence different H_{m0} values, to be pooled into a single statistical sample independently of the specific wave height tested.

The exceedance probability of a dimensionless volume V^* follows a two-parameter Weibull distribution:

$$P_V(V^* > x) = \exp \left[- \left(\frac{x}{a} \right)^b \right], \quad (15)$$

where a is the *scale parameter*: it sets the characteristic magnitude of individual volumes and is directly related to the ratio q/N_{ow} (mean discharge normalised by the number of overtopping waves). The parameter b is the *shape parameter*: it controls the heaviness of the distribution tail, that is, the structure's propensity to generate disproportionately large events relative to the mean. A low value of b indicates a heavy tail and therefore a high risk of extreme volumes, even when a , and consequently q , remains moderate.

9.3. Inverting the Equation to Estimate $V_{P\%}$ and $V_{\max}$

To use the fitted distribution, Equation 15 is inverted to obtain the volume associated with a given exceedance probability $P_{V\%}$:

$$V_{P\%}^* = a \left[-\ln \left(\frac{P_{V\%}}{100} \right) \right]^{1/b}. \quad (16)$$

The expected maximum individual volume for a sea state containing N_{ow} overtopping waves is obtained by setting the exceedance probability to $1/N_{ow}$:

$$V_{\max}^* = a (\ln N_{ow})^{1/b}, \quad (17)$$

with the dimensional volume recovered via $V_{\max} = V_{\max}^* H_{m0}^3$.

9.4. Implementation

For each configuration studied, the pooled sample of dimensionless volumes V^* from all sea states is fitted by maximum likelihood according to Equation 15, yielding the configuration-specific parameters (a, b) . Equations 16 and 17 are then applied to these parameters to estimate any desired exceedance volume and the expected maximum individual volume, respectively. The latter is directly comparable to the structural and safety criteria specified by EurOtop.

9.5. Results and Discussion

The results of the regression of the Weibull parameters a and b against the Construction Completion Index I_h are shown in Figure 13. The regression demonstrates a clear relationship between the degree of construction completion and the statistical characteristics of individual overtopping volumes. As I_h increases, indicating a more complete structure, both the scale parameter a and the shape parameter b tend to decrease, suggesting that the overtopping volumes are smaller and less variable.

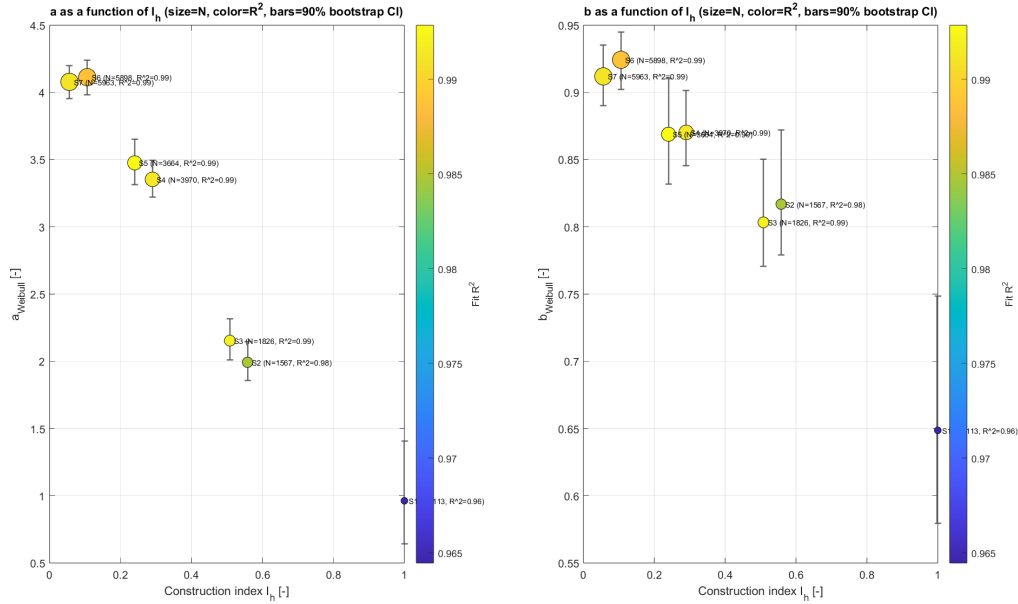


Figure 13: Regression of Weibull parameters a and b against the Construction Completion Index I_h .

The vertical bars represent the 90% bootstrap confidence intervals for the Weibull parameters (a and b). They illustrate the statistical uncertainty and variability of the fitted parameters for each structural degradation state. States with a low number of overtopping events, such as S_1 , exhibit wider confidence intervals because of their smaller sample sizes, whereas states with thousands of events yield much narrower and more reliable parameter estimates. Consequently, to ensure reliable predictions of $V_{\max}$, it is recommended to conduct a sufficient number of simulations for each construction stage, particularly for stages in which overtopping is infrequent.

10. Engineering Applications

The results obtained using the modified EurOtop equation with the new correction factors provide a first estimate of overtopping discharge during the construction phase of a rubble mound breakwater. The main purpose of this study is to present the general methodology and encourage further research. With improvements to the correction factors, the method could provide information for assessing overtopping risk during construction and planning construction activities accordingly. For example, each day, the person responsible for the construction could check the forecast sea state and use the modified EurOtop equation to answer the following questions:

- Is overtopping expected tomorrow?
- How many overtopping events are expected?
- What is the largest overtopping discharge that may occur during the day?
- Is this largest discharge acceptable for the construction activities?
- If not, what damage could occur and what would the consequences be?

11. Limitations and future Research

Two-Dimensional Simulations

The simulations were performed in two dimensions, which may not fully capture the three-dimensional effects of wave interactions with the breakwater. Future research could involve three-dimensional simulations and include

the parameter γ_β to account for wave-obliquity effects. This would provide a more comprehensive understanding of overtopping behaviour during construction.

Single Breakwater Configuration

The simulations were conducted for a single rubble mound breakwater configuration, which may limit the generalisability of the findings. Future studies could explore a wider range of breakwater configurations, including different slopes, crest heights, and layer compositions, to validate the applicability of the modified EurOtop equation across diverse scenarios.

Absence of Crown Walls

The simulations did not include crown walls, which can influence overtopping behaviour. Future research could investigate the impact of crown walls and other structural elements that may act as the final barrier to waves and significantly influence overtopping discharge during construction.

Recommendations for Future Work

To improve understanding of overtopping discharge during the construction phase of a breakwater, the following recommendations are proposed:

- Conduct three-dimensional simulations to capture the full complexity of wave interactions with the breakwater.
- Explore a wider range of breakwater configurations to validate the modified EurOtop equation.
- Investigate the impact of crown walls and other structural elements on overtopping behaviour.
- Consider the effects of wave obliquity and other environmental factors on overtopping.
- Use another CFD model to validate the results obtained with the IH2VOF model and assess the robustness of the findings.
- Conduct additional simulations under different wave conditions to validate the modified EurOtop equation.

- Explore alternative methods for correcting the EurOtop equation and introduce new parameters to account more accurately for the construction phases of a breakwater.
- Investigate the estimation of the maximum overtopping volume $V_{\max}$ and its implications for structural safety and risk management during construction.

12. Conclusions

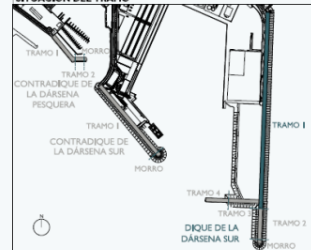
This internship was an enriching experience, both professionally and personally. I learned a great deal about coastal engineering and the challenges associated with the design and construction of breakwaters. I also learned how to use a CFD software developed by a public organization. Although the results are not yet complete, the methodology developed during this internship can be improved and adapted to different scenarios.

I also gained valuable experience in analysing complex datasets and writing a scientific paper. I had to write an abstract to submit my paper to the ICE breakwater 2027 congress, this was an enriching experiment. This experience confirmed my decision to pursue a career in industry rather than research. Working independently on this project allowed me to develop my problem-solving skills and become more self-reliant. However, I found it difficult to work alone on a project and would have appreciated more interaction with other people.

Appendix A. Construction Geometries

First Section

SITUACIÓN DEL TRAMO	
1	2
3	4
5	6
7	8
9	10
11	12
13	14
15	16
17	18
19	20
21	22
23	24
25	26
27	28
29	30
31	32
33	34
35	36
37	38
39	40
41	42
43	44
45	46
47	48
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59	60
61	62
63	64
65	66
67	68
69	70
71	72
73	74
75	76
77	78
79	80
81	82
83	84
85	86
87	88
89	90
91	92
93	94
95	96
97	98
99	100



DATOS GEOMÉTRICOS	
Longitud del tramo correspondiente a la sección tipo (m)	918
Profundidad mínima del fondo marino respecto a la BMVH (m)	13
Profundidad mínima del fondo marino respecto a la BMVH (m)	13
DATOS GEOTÉCNICOS	
Naturaleza del fondo marino	Arenas
CRITERIOS DE DISEÑO	
Vida útil (años)	25
Periodo de retorno T (años)	112
Criterios de inversión	Inicio de vertido
Altura de ola significativa H ₀₃ (m)	4,37
Periodo de pico T _p (s)	13
Incidencia de oleaje	Oblícuo
Altura de ola de diseño H _d (m)	5,55
MÉTODOS DE VERIFICACIÓN DE LA SECCIÓN	
Fórmulas matemáticas	X
Métodos numéricos	X
Ensayos en modelo físico	X

DATOS DEL MANTO PRINCIPAL		
Tipo de pieza		Bloques
Peso (t)	20,00	
Dimensiones labral (m)		2,00x2,00x0,20
Peso específico (dm ³)	2,10	
Talud		1,5x1/4
Nº de ripas	2	
Cota superior del manto (m)	5,91	
Anchura en coronación del manto (m)	4,70	
Cota inferior del manto (m)	4,00	
DATOS DE LA BÉRNIA DE APOYO DEL MANTO PRINCIPAL		
Tipo de la pieza		Escolleros
Peso (t)	3,00	
Talud		1,5x1/4
Cota de coronación (m)	4,00	
Anchura de coronación (m)	3,15	
DATOS DEL ESPALDÓN		
Cota de coronación (m)	7,50	
Cota de cimentación (m)	0,48	
Existencia de material sustitutable, según se trate de:		

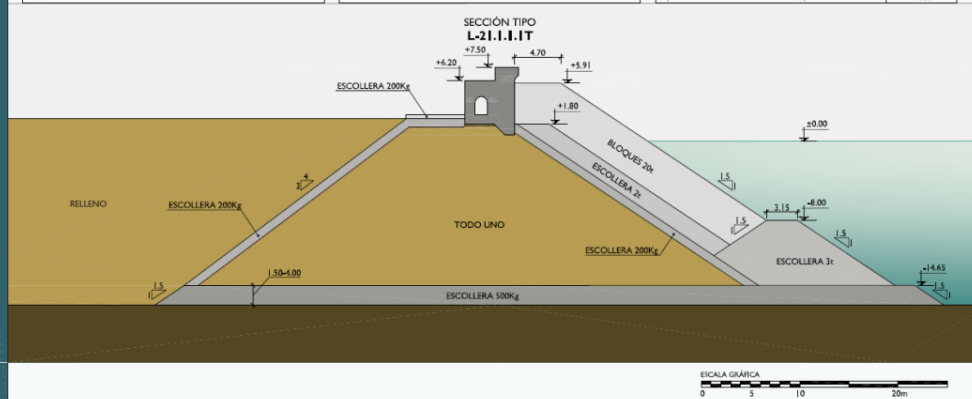


Figure A.14: Original section

Second Section

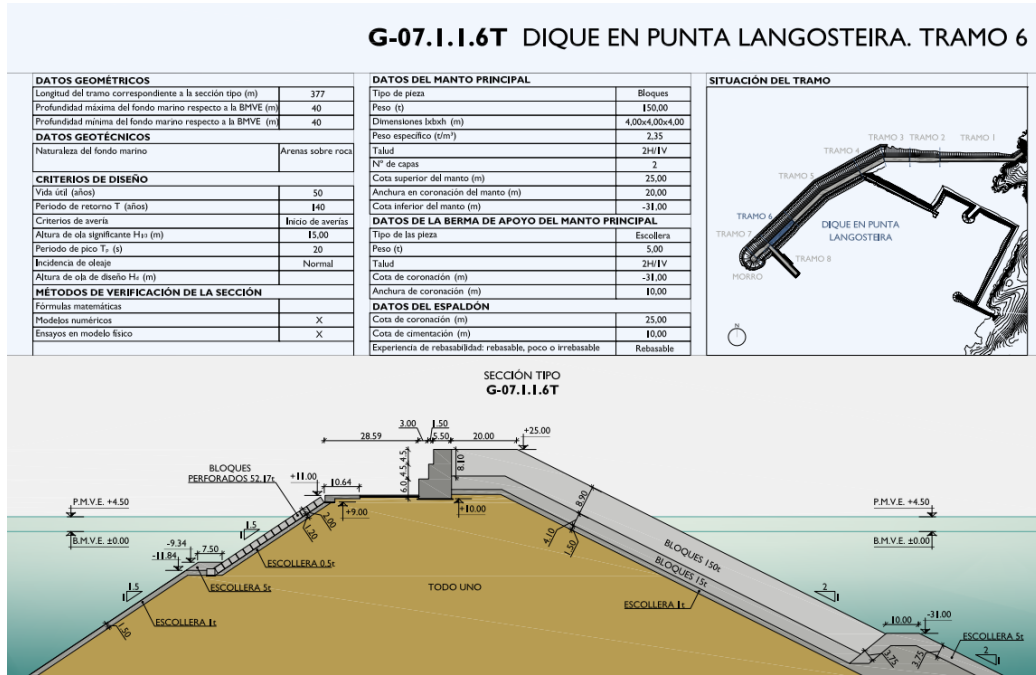


Figure A.15: Independent geometry for validation purposes.

Appendix B. Calibration Coefficients

The following table summarises the calibration coefficients and weighting factors used in the modified EurOtop equation to predict overtopping discharge during the construction phase of rubble mound breakwaters.

Table B.4: Calibration Coefficients and Weighting Factors

Parameter	Section 1
α (Calibration constant)	7.36
β (Calibration exponent)	2.71
w_R (Crest freeboard weight)	0.70
w_A (Armor layer weight)	0.15
w_{U1} (Underlayer 1 weight)	0.10
w_{U2} (Underlayer 2 weight)	0.05

Fiche d'évaluation étudiant

Ce document est à insérer dans le rapport. Le rapport est à fournir pour le 24 aout 2026 dernier délai.

Organisme d'accueil : IHCantabria

Sujet du stage : Modification of the Eurotop overtopping equations for breakwaters during their construction

Evaluation qualitative du stage	I	S	B	E	N	Commentaires
Qualité de l'entreprise						L'institut embauche une ancienne élève
L'entreprise offre-t-elle un contexte propice à une carrière d'ingénieur ?				X		
L'entreprise connaissait elle l'école (accueil de stagiaire, embauche, relations autres...) ?			X			
L'entreprise a-t-elle mis à votre dispositions les moyens nécessaires pour réaliser votre mission (documents, éléments d'information, matériels) ?				X		
Qualité de la missions						
Vos missions étaient-elles en rapport avec votre formation ?			X			
Les missions effectuées étaient-elles bien celles définies au départ ?				X		
Qualité de l'encadrement						J'ai travaillé seul 95% du temps.
Votre tuteur organisme d'accueil a-t-il pris le temps de vous présenter le fonctionnement de la structure et l'équipe ?			X			
Votre tuteur organisme d'accueil vous a-t-il aidé et conseillé quand cela était nécessaire ?		X				
Votre enseignant référent vous a-t-il aidé et conseillé lorsque cela était nécessaire ?			X			

Explication des cotations	
I	(I) Insuffisant
S	(S) Suffisant
B	(B) Bien
E	(E) Excellent
N	(N) Non applicable

NB : Dans le but d'alléger la lecture du document, le genre masculin est utilisé sans discrimination pour le genre masculin et féminin.

Fiche d'évaluation étudiant

Autoévaluation : développement des compétences et trajectoire professionnelle

En prenant un peu de recul sur votre activité durant le stage pensez-vous avoir travaillé / développé certaines des compétences du référentiel de la formation Seatech ? Lesquelles ? Pourquoi et comment ? D'autres compétences ?

Observations sur les critères :

 Ces critères doivent être développés, même partiellement, au cours du stage de 2A.

 Utiliser, autant que possible, les mises en situation du stage de 2A pour consolider ces critères déjà développés en stage 1A

Compétence	Critère	Commentaire
Concevoir des solutions d'ingénierie	Analyser le besoin	J'ai passé un temps important à comprendre le besoin de corriger les équations Eurotop.
	Répondre au besoin	J'ai apporté une modification à ces équations. Bien qu'elle ne soit pas parfaite, j'ai expliqué les limites de son application.
	Utiliser les outils appropriés	J'ai utilisé les logiciels que m'a conseillé mon maître de stage.
	Documenter les choix, les sources et les références	J'ai cité toutes mes sources hormis les documents privés et les manuels internes de l'institut.
Mettre en œuvre des solutions	Analyser et améliorer une solution existante	J'ai passé un temps important à m'approprier les concepts et les outils et j'ai ensuite pu développer une méthodologie pour les améliorer après avoir compris le besoin
Développer une démarche R&D	Réaliser une veille technologique / un état de l'art	J'ai analysé toutes les solutions existantes du domaine y compris les solutions en cours de développement à savoir ; les méthodes de prévisions par réseaux de neurones.
	Formuler des hypothèses	J'ai fait des hypothèses simplificatrices de mon brise vague pour rendre le sujet plus global.
Piloter des projets d'ingénierie	S'inscrire dans le pilotage d'un projet	J'ai piloté le projet car j'étais tout seul et j'ai dû prendre mes propres initiatives.
	Prendre en compte la gestion globale des organisations et les règles économiques et juridiques	J'ai respecté les règles de l'institut.
	Utiliser les outils de gestion de projets et collaboratifs	

Fiche d'évaluation étudiant

Encadrer une équipe	S'insérer et collaborer	Je n'ai collaboré qu'avec mon maitre de stage.
	Travailler en équipe pluridisciplinaire et/ou internationale	J'ai communiqué en anglais avec mon maitre de stage.
	Communiquer (écrit et oral) de manière adaptée	J'ai rédigé un papier scientifique.
Agir en professionnel responsable	Prendre en compte les enjeux éthiques et sociétaux (RSE, DD, RGPD, ...)	
	Assumer la responsabilité de ses actes et décision	J'ai formulé des hypothèses parfois trop simplificatrices qui m'ont obligées à revenir de plusieurs jours/ semaines en arrière.
	Être dans une dynamique d'apprentissage	J'ai toujours cherché à apprendre des nouvelles méthodologies ou de nouvelles méthodes statistiques pour corriger une équation.
Trajectoire professionnelle A la suite de votre stage, avez-vous confirmé ou affiné votre projet professionnel d'être ingénieur (métier plus précis, secteur, contexte ou type d'entreprise, ...) ? Si oui, quelles actions pensez-vous devoir entreprendre pour y arriver (renforcer certaines connaissances, développer certaines compétences, lesquelles)? Je veux être ingénieur dans l'industrie, la recherche ne me permet pas de m'épanouir pleinement car on est souvent bloqué pendant des semaines sans solutions et le fait d'être bloqué seul est encore plus frustrant. Cependant j'ai découvert que le secteur de la protection des cotes m'intéresse et je n'exclue pas la possibilité de me spécialiser dans ce domaine. Date et signature de l'étudiant :		

Le 14/08/2026



To be filled in by the host organization tutor *and returned by August 24th 2026* to ecole-entreprise@seatech.fr

Host organization: **Environmental Hydraulics institute of the University of Cantabria (IHCantabria / UC)**

Student: **Quentin LHommeau**

Subject of internship: **Extended EUROTOP equation 6.5 for the rubble mound breakwaters during construction**

Assessment: Skill development

The evaluation grids is consistent with the study skills framework. The 6 skills are developed over the three years of study. Please evaluate the student according to the different criteria.

Reference for cotation

(A) Excellent	Beyond expectations (beyond students of the same level)
(B) Suitable	Meets expectations (within the norm for students at the same level)
(C) Insufficient	Does not meet expectations (below students at the same level)
(N) Not applicable	Criterion not assessable in the context of the internship

Observations on the criteria



These criteria must be developed, even partially, during the second-year internship.



Use as much as possible, practical situations of second-year internship to consolidate these criteria already developed in first-year internship.

Skill	Criterion	Assessment	Comment
Come up with engineering solutions	Analyse need	A	Quentin demonstrated a very good understanding of the engineering problem and was able to identify the main physical and numerical aspects governing wave overtopping during breakwater construction.
	Answer need	A	He successfully translated the engineering problem into a structured numerical methodology, proposing practical approaches to improve overtopping prediction during intermediate construction stages.
	Use appropriate tools	A	He showed very good ability in the use of numerical modelling and data-processing tools, particularly IH2VOF and MATLAB, independently handling a significant number of simulations and their post-processing.
	Document choices, sources and references.	A	The methodological choices were generally well justified and supported by relevant technical references, particularly the EurOtop methodology and previous overtopping research.
Implement solution	Analyse existing solution	A	Quentin critically analysed the existing EurOtop formulations, identifying their limitations when applied to partially constructed rubble-mound breakwaters.
	Propose a new solution	A	He proposed and implemented an original correction methodology based on the construction stage of the breakwater, including calibration and independent validation procedures.
	Adhere the principles of continuous improvement	A	Throughout the internship, he progressively refined the methodology based on the results obtained, testing alternative formulations and additional geometries.

	Write a scientific document and technical support	B	The work has been structured as a scientific paper and contains the main elements required to communicate the methodology and results. Some further improvement is still needed in the scientific writing, synthesis and discussion of the results.
--	---	---	---

Develop R&D steps	Carry out continuous technological surveillance	A	Quentin reviewed and applied current methodologies for wave overtopping prediction and numerical modelling, appropriately positioning his work within the existing state of the art.
	Formulate hypotheses.	A	He demonstrated a strong ability to formulate and test hypotheses regarding the influence of breakwater construction stages, geometry and layer completion on overtopping discharge.
	Propose an experimental approach, a protocol or a model.	B	He developed a consistent numerical methodology and simulation protocol. The approach is appropriate, although further work is required to extend and fully validate the proposed formulation for a wider range of configurations.
	Adopt an innovative approach	A	The introduction of a construction completion index and associated correction factors represents an innovative approach to addressing overtopping during intermediate construction stages.
Manage engineering projects	Get involved in project management	N	
	Take into account the overall management of organizations or the operating, economic or legal	N	
	Use project management and collaborative tools	A	Quentin worked in an organised and autonomous manner, effectively managing the numerical simulations, data processing and progressive development of the research tasks.
	Identify or mobilize resources appropriate	N	
Coach a team	Integrate and collaborate	N	
	Team working multidisciplinary and/or international	N	
	Communicate (written and oral) in adapted manner	A	He communicated his progress and technical difficulties clearly during the internship and was receptive to discussion, feedback and proposed modifications.
	Ensure leadership animation	N	
Act like a responsible professional	Take into account ethical and societal issues	N	
	Take responsibility for his actions and decision	A	Quentin showed a high level of autonomy and responsibility in carrying out the assigned work and in making technical decisions throughout the internship.
	Being open to learning new things.	A	He showed excellent willingness to learn new numerical and engineering concepts and was able to rapidly incorporate them into his work.
	Take a critical look at the meaning of the activity carried out	A	He demonstrated good critical thinking, identifying limitations in the proposed methodology and recognising cases where the calibration or validation requires further investigation.

Overall assessment of the intern and recommendations for occupying an engineering position (it is recommended to discuss this evolution with the intern).

Quentin has completed a very successful internship at IHCantabria. He has demonstrated a high level of autonomy, technical competence and motivation, particularly in numerical modelling, data analysis and coastal engineering. He was able to address a complex research problem, develop and test an original methodology, and critically analyse both its performance and limitations. He has also shown a very positive attitude towards learning and incorporating feedback throughout the internship. As a direct outcome of the work carried out during his internship, the results will be presented at the ICE 2027 *Coasts, Marine Structures and Breakwaters* Conference, which further demonstrates the scientific and technical relevance of the work developed. Overall, his performance has been excellent for a student at this stage of his education, and I consider that he has the technical and personal qualities required to successfully progress towards an engineering and R&D position.

Date, stamp of the host organization, name and signature of the tutor:

DIAZ HERNANDEZ
GABRIEL - DNI
72209734S

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DIAZ HERNANDEZ GABRIEL
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